

	Course	Databases and Information Systems			2026
	Exercise Sheet	1			
	Points	–			
	Release Date	April 28th 2026	Due Date	May 11th 2026	

Exercise Sheet 1: Relational Databases

With this exercise sheet, you prepare a part of the demo to be presented at the end of the semester. This requires working practically. Unfortunately, our exercise rooms this year are not equipped with PCs and the PC pools are not entirely usable, yet, so please bring a laptop or tablet if you want to work in the exercise rooms or need to show something to explain a question. If you do not have any portable device you can use for the exercise, please consider using the lending service of the General Students' Committee¹.

Please read the whole exercise sheet before starting to work. Take notes after every executed step and save your source code for being able to answer the questionnaire in section 3. All questions must be answered reasonably to pass the exercise. However, the answers are not graded as they are highly individual to the setup you are using.

1 Find a partner

Find one or two partners to form a team of 2 or 3 people. You may use the forum 'Team matchmaking'² which is pinned to this week's block in moodle if you can not find a partner offline. This team should stay the same for the remainder of the semester. When you found a partner, navigate to this week's material in the moodle course and find the activity 'Demo Teams'³. Create a new group or join an already existing group. You can chose the name of your group and add a description. If you want to have control over who joins your group, there is also the possibility to protect your group with a password.

2 Set up and use a relational database system

a) Set up a database system of your choice. Make sure it provides the following features:

- Provide a defined isolation level or at least claims to do so, unregarded of the specific label the system puts on it
- Ability to connect multiple clients to the same database using a programming language of your choice
- Ability to define and run transactions
- (Semi)automatically fill a table from data in a csv file

¹<https://www.asta-uhh.de/4-service/1-verleihservice.html>, accessed 15/04/2026

²<https://lernen.min.uni-hamburg.de/mod/forum/view.php?id=277740>

³<https://lernen.min.uni-hamburg.de/mod/groupselect/view.php?id=277561>

	Course	Databases and Information Systems			2026
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- b) This step provides an impression of how your system treats concurrent transactions from different connections. You are free to use any programming language of your choice. However, the level of support we can provide decreases depending on how exotic your language of choice is. Execute the following steps **exactly in the prescribed order**:
- i) Create a new persistent database, i.e., no main memory database, and connect to it. keep the connection open (connection A). **Do not use a graphical user interface!**
 - ii) Create a new table in your database and insert a tuple.
 - iii) Establish a second connection to the same database that was just created in step i (connection B).
 - iv) Start a new transaction in each of the two connections.
 - v) Use connection A to add a new record to the table created in step ii.
 - vi) Use connection B to add another new record to the table created in step ii. Make sure the values are not the same as in connection A.
 - vii) Check the contents of the table in both connections.
 - viii) Commit the transactions in both connections. In case of success, check the contents of the table in the successful connection(s).
 - ix) Only in case both transactions committed successfully: Open a third connection and check the contents of the table.
 - x) Start a new transaction in each connection.
 - xi) Modify the record created in step ii in each transaction with an UPDATE statement. Make sure the updated values are different from each other and from the existing value.
 - xii) Commit both transactions. In case of success, check the contents of the table in the successful connection(s).
 - xiii) Only in case both transactions committed successfully: Open a third connection and check the contents of the table.
- c) In this step, data from an external source is read into a database including a few adjustments. This is also a little recap of some topics covered in the foundations lecture. Start with a fresh database using only one connection, and follow the steps below:
- i) Download the 'Coffee Sales' dataset from moodle. This is a csv file containing data about coffee sales.
 - ii) Identify constraints which are useful in a database table but not provided by the csv file, e.g., key constraints and conditions a column entry might have to meet.

	Course	Databases and Information Systems			2026
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- iii) Create a new table and insert the data from the csv file as automatically as possible. Define the data types yourself or check the automatically detected types after reading the file. Check if the data is really there after insertion.
- iv) If not already done during the creation of the table, add the constraints identified in step ii. If necessary, add a new column for the primary key.
- v) Is there redundant information in the table? If yes, normalize it, i.e., store the redundant information in a different table and define a foreign key relation.

3 Questionnaire

Go to moodle and answer the questionnaire 'Questionnaire for Exercise Sheet 1'⁴. The questionnaire can only be answered when the team building element in moodle ('Demo Teams'⁵) is marked as done by you.

⁴<https://lernen.min.uni-hamburg.de/mod/feedback/view.php?id=277757>

⁵<https://lernen.min.uni-hamburg.de/mod/groupselect/view.php?id=277561>